

MAT1002 Lecture 14

Outline

- Partial derivatives
- Differentiability

Partial Derivatives

Definition

More generally, if $f : D \rightarrow \mathbb{R}$ is a function with $D \subseteq \mathbb{R}^n$, and

$$\vec{a} = \langle a_1, a_2, \dots, a_n \rangle \in D,$$

then the **partial derivative** of f at $\vec{a}$ with respect to the i -th coordinate (or i -th variable), denoted by $\frac{\partial}{\partial x_i} f(\vec{a})$, is defined to be the following limit:

$$\lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_{i-1}, a_i + h, a_{i+1}, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h}.$$

e.g. $\frac{\partial}{\partial z} f(a, b, c) = \lim_{h \rightarrow 0} \frac{f(a, b, c+h) - f(a, b, c)}{h}.$

To compute the partial derivative of f with respect to the i -th variable x_i , just treat all other variables as constants and differentiate f with respect to x_i .

Example

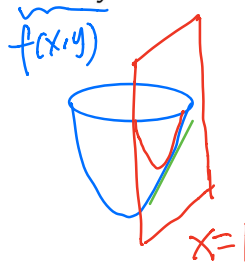
$$f_x(x, y) = 2x + 3y, \quad f_y(x, y) = 3x + 1$$

(a) For the function defined by $f(x, y) := x^2 + 3xy + y - 1$, find

$$\frac{\partial}{\partial x} f(4, -5) \quad \text{and} \quad \frac{\partial}{\partial y} f(4, -5).$$

$= -7 \qquad \qquad \qquad = 13$

(b) Find the slope of tangent to the parabola, obtained by intersecting the surface $z = x^2 + y^2$ with the plane $x = 1$, at the point $(1, 2, 5)$.



• Parabola is $g(y)$
 $z = f(1, y) = 1 + y^2.$

• Slope of tangent is

$$\left. \frac{dg}{dy} \right|_{y=2} = \left. \frac{\partial f}{\partial y}(1, y) \right|_{y=2} = 2y \big|_{(x,y)=(1,2)} = 4.$$

If f is a function with two variables, then f_x and f_y are also functions with two variables, so they can have partial derivatives. These are the second-order partial derivatives of f , whose notations are given as follows:

$$\frac{\partial^2 f}{\partial x^2} := f_{xx} := (f_x)_x = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right),$$

$$\frac{\partial^2 f}{\partial y \partial x} := f_{xy} := (f_x)_y = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right),$$

$$\frac{\partial^2 f}{\partial x \partial y} := f_{yx} := (f_y)_x = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right),$$

$$\frac{\partial^2 f}{\partial y^2} := f_{yy} := (f_y)_y = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right).$$

e.g. $\frac{\partial^3 f}{\partial x \partial y \partial z} = f_{zyx}$



These notations extend to functions with three or more variables.

Exercise

Compute the four second-order derivatives of the function f defined by

$$f(x, y) := x \cos y + ye^x.$$

In the exercise above, it turns out that $f_{xy} = f_{yx}$. This is not a coincidence.

Theorem (Mixed Derivative Theorem) *disk/ neighbourhood*

Suppose that f is defined on an open ~~ball~~ B containing a point (a, b) . If f_{xy} and f_{yx} are continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Example

Find $\frac{\partial^2 w}{\partial x \partial y}$ if $w = xy + e^y(y^2 + 1)^{-1}$.

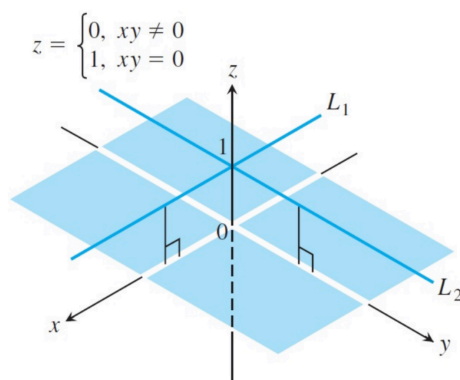
Ans : 1.

The Mixed Derivative Theorem extends to functions with more variables and higher order derivatives, as long as all the derivatives of this order are continuous, e.g.,

$$f_{xyz} = f_{yzx} = f_{zxy} = \dots$$

Existence of Partial Derivatives Does Not Imply Continuity

Even if all the partial derivatives of a function f exists at a point, it does not mean that the function is continuous at that point. One example is shown in the following figure.



Another example is given below.

Example

The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := \begin{cases} \frac{xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

has both partial derivatives at $(0, 0)$. However, it is not continuous at $(0, 0)$, since we have shown that

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$$

does not exist.

It turns out that just having all partial derivatives at a point does not mean that the function is differentiable at that point.

Differentiability

Consider a one-variable function $y = f(x)$, and suppose that f is differentiable at x_0 . Then ^{fixed}

$$\varepsilon := \varepsilon(\Delta x) = \frac{\overbrace{f(x_0 + \Delta x) - (f(x_0) + f'(x_0)\Delta x)}^{\text{Error of lin. approximation}}}{\Delta x} \rightarrow 0$$

as $\Delta x \rightarrow 0$. In other words, there exists $\varepsilon := \varepsilon(\Delta x)$ s.t.

$$\underline{\Delta y = f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + \varepsilon\Delta x}, \text{ where}$$

$$\varepsilon \rightarrow 0 \text{ as } \Delta x \rightarrow 0. \quad (*)$$

(*) is true essentially because tangent line approximation is "good" at $x = x_0$.

Intuitively, we want a two-variable function f to be differentiable at (x_0, y_0) if the graph of f has a tangent plane at (x_0, y_0) . Formally, the definition can be extended from (*).

Def: A function $z=f(x,y)$ is **differentiable** at $(x_0, y_0) \in D$ if both f_x and f_y exist at (x_0, y_0) , and

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

satisfies

$$\Delta z = f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

for some $\varepsilon_1 := \varepsilon_1(\Delta x, \Delta y)$ and $\varepsilon_2 := \varepsilon_2(\Delta x, \Delta y)$ such that $\varepsilon_1 \rightarrow 0$ and $\varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$. We

say f is **differentiable** on D if f is differentiable at all points in D .

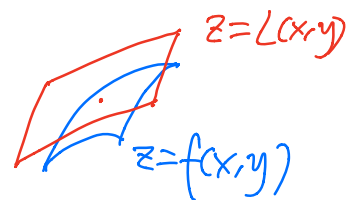
Same as $\sqrt{(\Delta x)^2 + (\Delta y)^2} \rightarrow 0$.

Q: What does it have to do with tangent plane?

- Note that if $z=L(x,y)$ the tangent plane to the graph of $z=f(x,y)$ at (x_0, y_0) , then by geometry, one can see that

$$L(x_0 + \Delta x, y_0 + \Delta y) = f(x_0, y_0) + f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y,$$

So



$$\begin{aligned}
& \Delta z - (f'_x(x_0, y_0)\Delta x + f'_y(x_0, y_0)\Delta y) \\
&= f(x_0 + \Delta x, y_0 + \Delta y) - (f(x_0, y_0) + f'_x(x_0, y_0)\Delta x + f'_y(x_0, y_0)\Delta y) \\
&= f(x_0 + \Delta x, y_0 + \Delta y) - L(x_0 + \Delta x, y_0 + \Delta y).
\end{aligned}$$

Hence, in the def. of differentiability, it says that the error of the tangent plane approximation should equal

$$\varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

for some ε_1 & ε_2 with ε_1 & $\varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

Exercise Show that if $E(\Delta x, \Delta y) = \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$ where ε_1 and ε_2 satisfy the condition above, then

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{E(\Delta x, \Delta y)}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = 0.$$

e.g. The function $f(x, y) = xy$ is differentiable at all (x_0, y_0) , since

$$\begin{aligned}
& f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) - f'_x(x_0, y_0)\Delta x - f'_y(x_0, y_0)\Delta y \\
&= (x_0 + \Delta x)(y_0 + \Delta y) - x_0 y_0 - y_0 \Delta x - x_0 \Delta y \\
&= \Delta x \Delta y = o(\Delta x) + o(\Delta y), \text{ where } \Delta x \rightarrow 0 \\
&\text{as } (\Delta x, \Delta y) \rightarrow (0, 0).
\end{aligned}$$

In practice, definition may be hard to use for determining differentiability.

The following is a sufficient condition for a two-variable function f to be differentiable at a point.

Theorem

disk/ neighbourhood

Suppose that partial derivatives f_x and f_y exist in an open ball containing (x_0, y_0) and are continuous at (x_0, y_0) . Then f is differentiable at (x_0, y_0) .

By this theorem, for $f(x, y) = xy$, since $f_x = y$ and $f_y = x$ are both continuous, we see that f is differentiable.

If f is differentiable at (x_0, y_0) , then as $(\Delta x, \Delta y) \rightarrow (0, 0)$, $\epsilon_1 \Delta x + \epsilon_2 \Delta y \rightarrow 0$, so $\Delta z \rightarrow 0$ by def, i.e.,

$$f(x_0 + \Delta x, y_0 + \Delta y) \rightarrow f(x_0, y_0).$$

In other words,

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} f(x_0 + \Delta x, y_0 + \Delta y) = f(x_0, y_0),$$

So f is continuous at (x_0, y_0) . This gives the "familiar" theorem.

Theorem If a two-variable function f is differentiable at (x_0, y_0) , then it is continuous at (x_0, y_0) .

Differentiability of a three variable function $w = f(x, y, z)$ is defined similarly: f is differentiable at (x_0, y_0, z_0) if f_x , f_y and f_z all exist at (x_0, y_0, z_0) , and

$$\Delta w = f_x(x_0, y_0, z_0)\Delta x + f_y(x_0, y_0, z_0)\Delta y + f_z(x_0, y_0, z_0)\Delta z + \varepsilon_1\Delta x + \varepsilon_2\Delta y + \varepsilon_3\Delta z$$

for some functions ε_1 , ε_2 and ε_3 (of Δx , Δy and Δz) satisfying $\varepsilon_i \rightarrow 0$ as $(\Delta x, \Delta y, \Delta z) \rightarrow (0, 0, 0)$, $\forall i \in \{1, 2, 3\}$.

The theorems above involving differentiability also hold for functions with three (or even more) variables.